

Rotating Snakes and Their Rainbow Powers: The More Colorful, the More Powerful

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Abstract

Visual illusions are often used to study the mechanisms of motion perception in the visual system. The present study examines the effect of perceived brightness and chromatic saturation on the intensity of Rotating Snakes illusions. A total of 36 college-aged participants viewed 40 trials of Rotating Snakes, comprising 5 trials of each condition, and responded with the rotation direction. Findings suggest that greater contrast between middle-luminance colors may strengthen illusory motion, potentially through perceived brightness differences associated with the Helmholtz-Kohlrausch effect. Colored Rotating Snakes produced stronger rotation, as indicated by shorter response times and higher response accuracy; the greyscale-2 condition produced weaker illusions than the greyscale-1 condition. The present findings also highlight limitations in current latency-based explanations of the Rotating Snakes illusion and motivate future research using more controlled methods and color-dependent variants of the Fraser-Wilcox illusion.

Introduction

The Rotating Snakes illusion is a variant of the Fraser-Wilcox illusion, made of rings with a striping pattern of four colors of different luminance levels, making the image appear to be rotating (Uesaki et al., 2024b). Such visual illusions are valuable in understanding and investigating the mechanisms underlying perception.

A commonly accepted explanation of the Rotating Snakes illusion is that the visual system responds faster to colors of higher contrast, thus creating an illusory movement in the direction from high to low contrast (Uesaki et al., 2024b). Conway (2005) tested this theory by recording the primary visual cortex (V1) and the middle temporal area (MT) activity of 3 macaque monkeys, and found supportive evidence demonstrating latency differences between response peaks to high contrast (black or white) and to low contrast (dark or light grey) colors.

While Conway (2005) did not find any significant effect of color, Uesaki et al. (2024b) report that Rotating Snakes illusions in blue-yellow color combination showed a stronger magnitude of illusory motion than those with red-green or greyscale color combinations. Uesaki et al. (2024b) had proposed chromatic saturation contrast as a potential explanation, since the chromatic saturation contrast between the blue and yellow elements in the BY condition was likely larger than the contrast between the red and green elements in the RG condition. However, the exact mechanism of how chromatic saturation creates illusory motions remained unclear, as previous research had emphasized luminance contrast as the main proponent of Rotating Snakes illusions. (Conway, 2005; Kitaoka, 2014b)

The Helmholtz-Kohlrausch (HK) effect is a visual phenomenon in which colors with greater chromatic saturation are perceived as brighter despite having equal physical luminance. (Corney et al., 2009) We consider this phenomenon relevant as the perceptual differences to

equiluminant stimuli are functionally represented in V1, where local motion signals composing the illusion are generated, such that V1 activity is better correlated with brightness than with physical luminance. (Uesaki et al., 2024b; Corney et al., 2009)

The present study attempts to test the hypothesis that the Rotating Snakes illusion depends on brightness contrast, rather than physical luminance contrast alone, by repeating the experiment done in Uesaki et al. (2024b) with an additional greyscale-2 condition to examine the effects of lower contrast between the middle luminance colors. We expect to replicate findings from Uesaki et al. (2024b), such that the blue and yellow condition produces the strongest illusion, followed by the red and green condition, then the greyscale conditions. Assuming this difference stems from the Helmholtz-Kohlrausch effect causing a larger perceived brightness contrast between the blue and yellow, we predict that the greyscale-2 condition should generate a weaker illusion than the original greyscale condition due to a reduced contrast between the greys.

Method

Participants

Our sample consisted of 36 subjects. All participants were college students taking part in a Lab in Perception Psychology course, and had no prior knowledge of Rotating Snakes illusions.

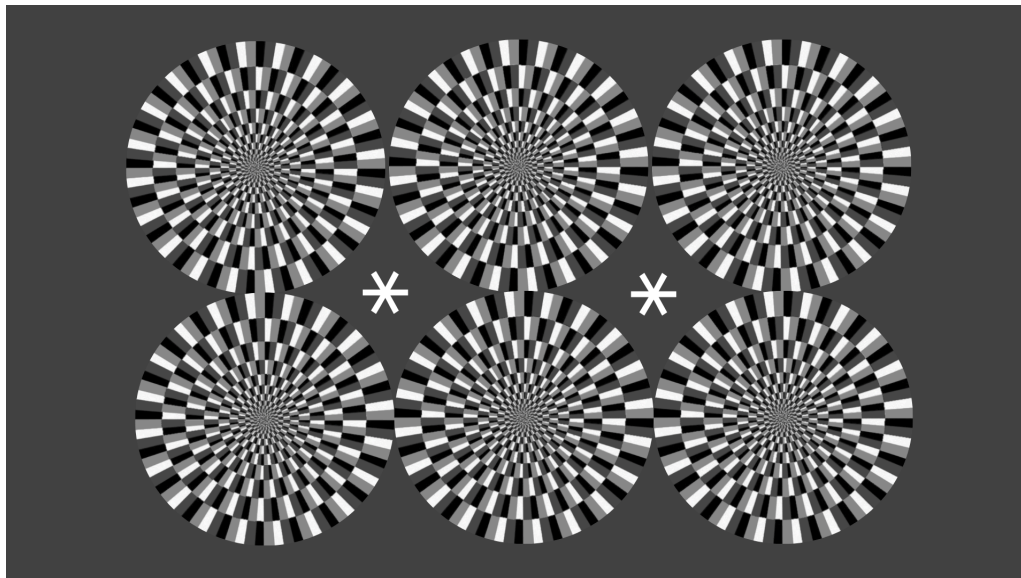
Stimuli

The Rotating Snakes illusions used for red-green, blue-yellow, and greyscale-1 conditions were taken from Uesaki et al. (2024b). To create greyscale-2, the greyscale-1 image was imported into Ibis Paint X on iPad Pro. The greys were then manually colored in such that dark grey was lightened (hex code #484848) and light grey was darkened (hex code #8A8A8A). In Canva, for each condition, 6 rotating snake circles were lined up in 2 rows of 3 snakes each, with

fixation stars at the 2 intersections of rows and columns, as shown in Figure 1. The backgrounds for all Rotating Snakes were modified to be lighter (hex code: #454545) so that background luminance was between the two greys in greyscale-2 and to ensure consistency across other conditions. For each color condition, one version had the color orders organized to create clockwise motion, while the other was manipulated to show counterclockwise motion.

Figure 1

Sample Greyscale-2 Clockwise Stimuli



Design

This study utilized a within-subjects design. Participants' accuracy and response times (RTs) in various color conditions were collected as a measure of illusion intensity. A one-way ANOVA was conducted on response times across color conditions. For planned contrasts, a paired t-test was conducted between each color condition to compare response times.

Apparatus

Participants were given a link on their personal laptops that directed them to their default web browser and launched the PsychoPy program via the Pavlov.org website. Throughout the

experiment, Rotating Snake images of 4 different color combinations with clockwise or counterclockwise rotation were randomly displayed on screen. Participants responded by pressing the arrow keys to indicate whether the illusory motion was clockwise, counterclockwise, or not moving.

Procedure

Text on screen instructed participants to press the left arrow key if they perceived the illusion rotating counterclockwise, the right arrow key if clockwise, or the down arrow key if the illusion didn't appear to be moving. It also informed them that there would be 40 trials in total. Participants then pressed space to begin the experiment.

A rotating snake stimulus is presented to the participant, and remains on screen until the participant responds. After which, the image is replaced with a blank screen with text instructing the participant to press the space bar to continue. This blank screen was added to prevent confounding illusory movements from switching between images.

This was repeated for a total of 40 trials; participants viewed each direction x color condition group 5 times in random order.

Results

Invalid data and responses with response times over 3999 ms or below 250 ms were removed before conducting the following analyses. Data from incorrect trials were also removed for response time analysis. An alpha level of 0.05 was used for all tests.

As seen in Figure 2, mean response times did not differ greatly across color conditions. However, mean reaction times were fastest in the BY condition ($M = 1434.27$ ms, $SD = 473.17$ ms), closely followed by the RG condition ($M = 1463.27$ ms, $SD = 433.27$ ms). Both greyscale conditions had slower responses, suggesting a weaker illusion, with greyscale-1 being

moderately faster than greyscale-2 (GS-1: $M = 1576.69$ ms, $SD = 629.82$ ms; GS-2: $M = 1635.58$ ms, $SD = 733.87$ ms).

Mean accuracy rates, pictured in Figure 3, can be seen to follow the same ranking order in reverse. The GS-2 condition had the lowest accuracy ($M = 0.63$, $SD = 0.33$), followed by the GS-1 condition ($M = 0.76$, $SD = 0.29$). Both colored conditions had higher accuracy rates by a noticeable margin, with the RG condition ($M = 0.91$, $SD = 0.16$) having a slightly lower response accuracy than the BY condition ($M = 0.95$, $SD = 0.14$).

A one-way within-subjects ANOVA on response time indicated a significant effect of the color condition on illusory rotation intensity ($F(3, 102) = 3.051$, $p < .05$). Of all six of the paired t-tests done on response time, only the BY and GS-2 conditions were found to have a significant difference from each other ($t = -2.553$, $p < .05$), the remaining t-tests did not show significance of color on response time, as seen in Table 1.

Table 1

T-Test Values for Response Time Over Color combinations

Conditions	Red-Green	GreyScale-1	GreyScale-2
Blue-Yellow	$t = -0.508$, $p = 0.6146$	$t = -1.9217$, $p = 0.06332$	$t = -2.553$, $p < .05$
Red-Green		$t = -1.6619$, $p = 0.106$	$t = -1.8081$, $p = 0.07945$
GreyScale-1			$t = 0.060242$, $p = 0.9523$

A second one-way within-subject ANOVA on accuracy suggested an even more significant effect of color combination on illusion strength ($F(3, 105) = 20.74$, $p < .001$). Paired t-tests revealed a significant effect between both colored conditions and both greyscale conditions (BY-GS-1: $t = 6.0552$, $p < .001$; BY-GS-2: $t = 4.6204$, $p < .001$; RG-GS-1: $t = 5.2537$,

$p < .001$; RG-GS-2: $t = 3.1552, p < .01$). There was also a significant difference between the two greyscale conditions ($t = -4.0412, p < .001$). All accuracy t-test results are displayed in Table 2.

Table 2

T-Test Values for Accuracy Over Color Combinations

Conditions	Red-Green	GreyScale-1	GreyScale-2
Blue-Yellow	$t = 1.4243, p = 0.1632$	$t = 6.0552, p < .001$	$t = 4.6204, p < .001$
Red-Green		$t = 5.2537, p < .001$	$t = 3.1552, p < .01$
GreyScale-1			$t = -4.0412, p < .001$

Like Uesaki et al. (2024b), both response time and accuracy results support the notion that the presence of color creates stronger illusory motions in Rotating Snakes. In addition, our results also imply that illusion is more intense when there is a stronger luminance or brightness contrast between the two middle luminance colors.

Figure 2

Response Times Across Color Conditions

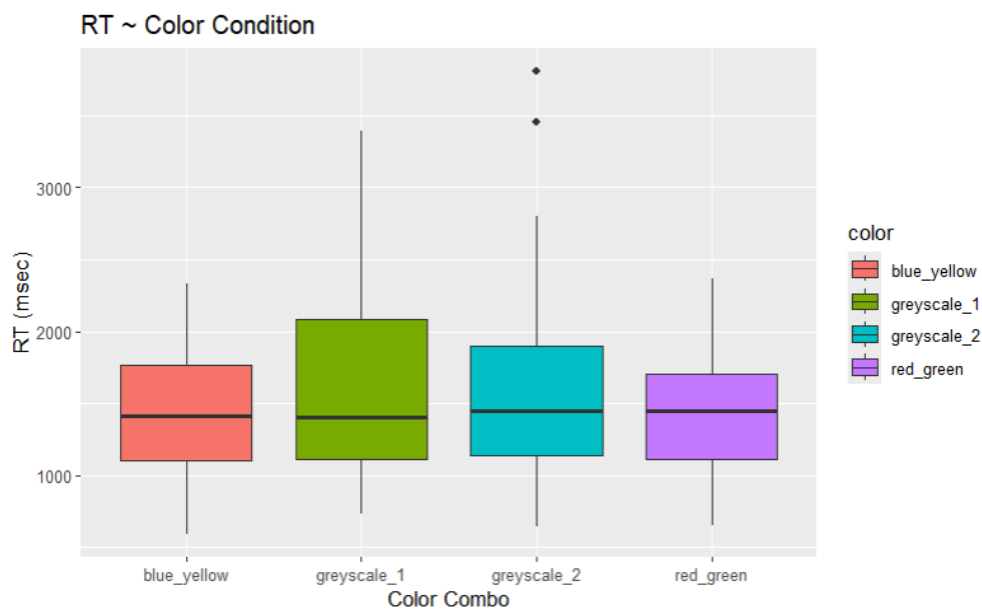
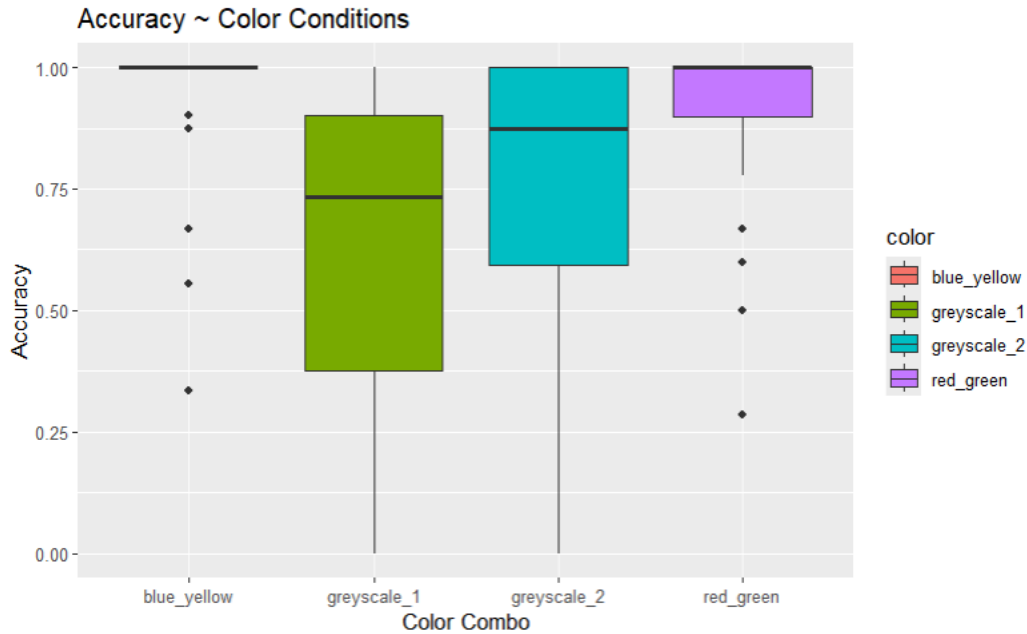


Figure 3*Mean Response Accuracy Across Color Conditions*

Discussion

The present findings provide evidence that a stronger contrast between middle-luminance colors also facilitates stronger illusory motion. The results are consistent with our interpretation based on Uesaki et al. (2024b), that the blue and yellow Rotating Snakes illusion, due to stronger chromatic contrast between the colors, had a larger contrast in perceived brightness and therefore creates a more intense illusion. Among the two grayscale conditions, grayscale-1 — which contained a larger contrast between the two middle grayscale values — produced significantly stronger illusory motion than grayscale-2, whose middle grayscale values were closer in luminance.

Since both colored conditions, despite the same physical luminance, are significantly more chromatically saturated than grayscale-1, the HK effect could explain the significantly

more robust illusion in the blue-yellow and red-green conditions compared to greyscale-1. Moreover, though Uesaki et al. (2024b) did not measure the chromatic saturation of colors, the potentially larger contrast in saturation might have led to a larger contrast in perceived brightness due to the HK effect, thus slightly intensifying the illusion in the BY condition.

At the same time, the present findings raise questions about the currently proposed mechanism underlying the Rotating Snakes illusion. Existing explanations suggest that the illusion arises from differences in response latency within the visual system to high- versus low-contrast regions. Under this framework, reducing contrast between the two middle-luminance colors should increase their contrast relative to the adjacent black and white regions, which would theoretically strengthen the illusion. However, this prediction was not supported in the present experiment, as grayscale-2 produced weaker illusory rotation instead. Future studies could therefore examine the black–dark gray and white–light gray pairings separately, similar to the approach used by Conway (2005), to better understand how local contrast relationships contribute to the illusion.

Methodological differences between the present and previous studies should also be considered. Uesaki et al. (2024b) quantified illusion strength using a method of constant stimuli, in which the images physically rotated at velocities ranging from -1.2 to 1.2 deg/s in increments of 0.3 deg/s. The velocity at which physical rotation cancelled out the perceived illusory motion was used as a measure of illusion intensity. In contrast, the present study relied on response time and accuracy measures due to technical limitations. Future research could replicate the present experiment, particularly the two grayscale conditions, using the method of constant stimuli to determine whether similar results emerge with a more direct measure of illusion strength. Additionally, Uesaki et al. (2024b) controlled for visual angle, as Rotating Snakes illusions are

known to be more salient in peripheral vision, whereas the present study did not. Repeating the experiment while controlling for visual angle may affect whether the present findings remain robust.

Further insight may also come from investigating color-dependent variants of the Fraser-Wilcox illusion. Kitaoka (2014b) proposed a color-dependent Fraser-Wilcox variation using four colors (dark purple, light purple, red, and pink) with only three luminance levels. In this illusion, the two medium-luminance colors differed in wavelength but not in brightness, and the illusion disappeared only when these colors were converted to grayscale. This finding suggests that wavelength differences themselves may contribute to illusory motion perception. Future work could use similar color-dependent rotating snake stimuli to isolate the effects of color saturation while minimizing luminance differences, allowing researchers to investigate the role of color independently from image luminance.

References

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